

Chapter 9 continued

- d** One mark for **one** correct, **one** mark for all correct and in the correct order.

12 151
13 28

- e** One mark for each completed statement.

```
SELECT SUM(Quantity bought)
FROM SALES
WHERE Date = 4/4/2020
```

- f** One mark for each completed statement.

```
SELECT SUM(Total Cost)
FROM SALES
```

Chapter 10: Boolean logic

Discussion questions

- 1 In Boolean, values are either true (1) or false (0). In a computer, if there is electricity that means 1 and if there is no flow of electricity that means 0.
- 2 A logic gate takes one or more inputs as a Boolean value (or values) and outputs a value.
- 3 All digital computers represent data in binary form with 0s and 1s, so they would not be able to operate without logic gates.

Questions

- 1 Input  Output

2

A	X
1	0
0	1

- 3 $X = \text{NOT } A$

- 4 Input 1  Input 2 Output

5

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

- 6 $X = A \text{ AND } B$

Chapter 10 continued



8

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

9 $X = A \text{ OR } B$



11

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

12 $X = A \text{ NAND } B$



14

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

15 $X = A \text{ NOR } B$



17

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

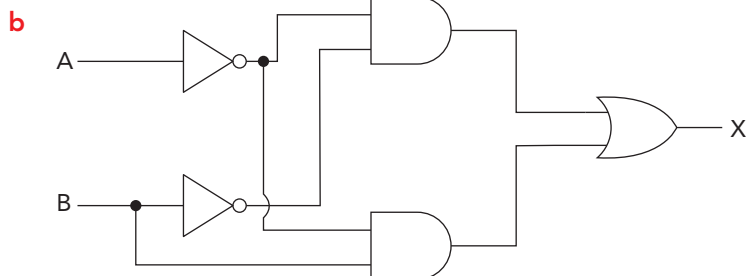
18 $X = A \text{ XOR } B$

19 $Q = (A \text{ OR } (\text{NOT } B \text{ AND } C))$

20 $X = ((A \text{ AND } B) \text{ OR } (A \text{ NAND } C))$

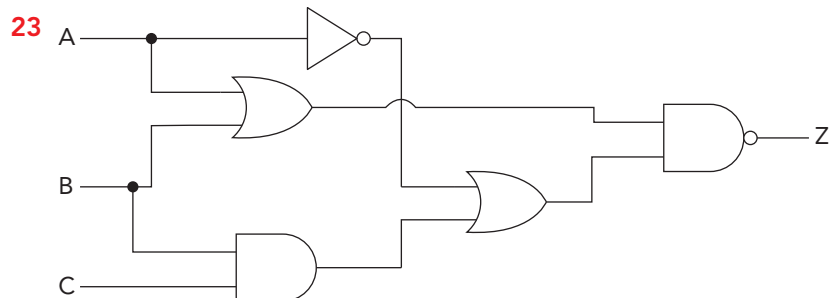
Chapter 10 continued

21 a $X = ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (\text{NOT } A \text{ AND } B))$



22

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



Skills focus 10.1 questions

1

A	B	C	Workspace	Q
0	0	0		0
0	0	1		1
0	1	0		0
0	1	1		0
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

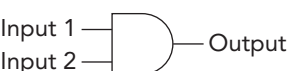
Chapter 10 continued

2

P	Q	R	Workspace	Z
0	0	0		0
0	0	1		0
0	1	0		0
0	1	1		1
1	0	0		0
1	0	1		0
1	1	0		0
1	1	1		0

Exam-style questions

1 To control the flow of electricity.

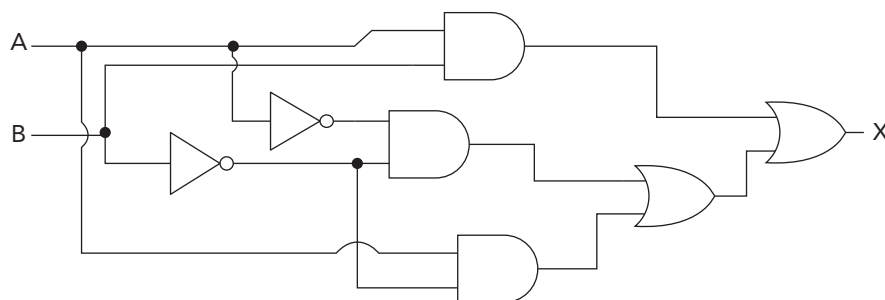
2 a 

b 

3 a **One** mark for each correct expression for each output (max **three**), **one** mark for joining each expression with OR.

$$X = ((\text{NOT } A \text{ AND NOT } B) \text{ OR } (A \text{ AND NOT } B) \text{ OR } (A \text{ AND } B))$$

b **One** mark for each correct AND gate with correct inputs.
One mark for both correct OR gates with correct inputs.



4 **One** mark for each correct expression for each output (max **four**), **one** mark for joining each expression with OR.

$$Z = ((\text{NOT } S \text{ AND NOT } T \text{ AND } P) \text{ OR } (\text{NOT } S \text{ AND } T \text{ AND NOT } P) \text{ OR } (S \text{ AND NOT } T \text{ AND NOT } P) \text{ OR } (S \text{ AND NOT } T \text{ AND } P))$$

5 a **One** mark for correct AND expression, **one** mark for correct NOR expression, **one** mark for joining these with NAND.

$$Y = ((\text{NOT } L \text{ AND } M) \text{ NAND } (L \text{ NOR } N))$$

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b Award:

Four marks for 8 correct outputs

Three marks for 6 or 7 correct outputs

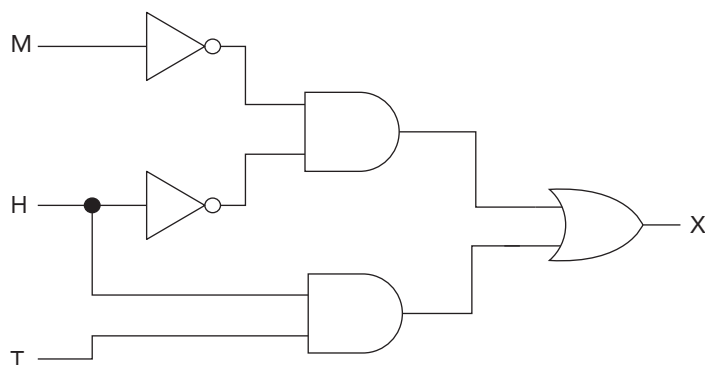
Two marks for 4 or 5 correct outputs

One mark for 2 or 3 correct outputs.

L	M	N	Workspace	Y
0	0	0		1
0	0	1		1
0	1	0		0
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		1
1	1	1		1

6 a $X = ((\text{NOT } M \text{ AND NOT } H) \text{ OR } (T \text{ AND } H))$

b **One** mark per each correct logic gate with correct inputs.



c

M	T	H	Workspace	X
0	0	0		1
0	0	1		1
0	1	0		0
0	1	1		1
1	0	0		0
1	0	1		0
1	1	0		0
1	1	1		1